# Kaua‘i – Tsunami ESF-6 playbook

**Data as of 2026-10-28**

**County:** Kaua‘i · **Hazard:** Tsunami

**Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.**

## 1. Summary

**Calculated by open code from public data · see Sources (section 8)**

| | | |
|---|---|---|
| People in the risk area<br>**6,161** | May need a public shelter (likely)<br>**308** | May need a public shelter (low – high)<br>**123 – 678** |
| Speak English less than well (C)<br>**139** | Use power-dependent medical equipment (estimate) (M)<br>**33** | People with a disability (I)<br>**610** |
| Age 65 and older (I)<br>**1,364** | People in households without a car (T)<br>**204** | Hearing difficulty (C)<br>**189** |
| Vision difficulty (C)<br>**118** | Cognitive difficulty (memory, decisions) (I)<br>**294** | Age 65 and older, living alone (S)<br>**271** |
| Known 2026 shelter sites usable for this hazard<br>**5** | People in the extreme tsunami zone (worst case)<br>**6,903** | |

Source: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER, FEMA National Risk Index v1.20; shelter rates: section 8.

Share of people in the risk area who may need a public shelter: low 2%, likely 5%, high 11%. One person can be counted in more than one need. Page 2 explains how to read these numbers.

## 2. Scope: county total vs. one community

**The county total is an upper bound.** It counts the people living inside the tsunami evacuation zone as hit at the same time. Real events usually hit fewer communities.

### Largest single community (the Census areas with the most people who may need a shelter)

| Community (Census area) | People in the risk area | May need a public shelter (low – likely – high) | Speak English less than well | Use power-dependent medical equipment (estimate) | People with a disability | Age 65 and older | People in households without a car |
|---|---|---|---|---|---|---|---|
| Kapa‘a<br>Census area 403.02 | 1,369 | 27 – **68** – 151 | 12 | 7 | 97 | 277 | 0 |
| Kekaha / Waimea<br>Census area 409 | 1,159 | 23 – **58** – 127 | 31 | 6 | 92 | 241 | 82 |
| Wailua / Wailua Homesteads<br>Census area 402.05 | 964 | 19 – **48** – 106 | 8 | 6 | 103 | 242 | 14 |

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

To plan for one community, use the first row.

## How to read this playbook

### Example row

| Community (Census area) | People in the risk area | May need a public shelter (low – likely – high) |
|---|---|---|
| Kapa‘a<br>Census area 403.02 | 1,369 | 27 – **68** – 151 |

Read it as: about 1,369 people live in the risk area of Kapa‘a. If this hazard hits, between 27 and 151 of them may need a public shelter; 68 is the likely number.

### Census area

A neighborhood-sized area the U.S. Census uses, usually 1,200–8,000 people. The number (for example "Census area 308") is its Census ID. Each area is named after the Census place where most of its people live. "near" means no one in the area lives inside a named place; "rural near" means most people live outside named places.

### Low – likely – high

The share of people in the risk area who may go to a public shelter: low 2%, likely 5%, high 11%. Where the rates come from: Team assumptions: likely 5%; low 2% (the hurricane planning factor); high 11% (the Lahaina 2023 rate).

### Who has access and functional needs (CMIST)

- **C – Communication:** people who speak English less than well (they may need interpreters and translated messages), and people with hearing or vision difficulty.
- **M – Medical:** people who use power-dependent medical equipment (an estimate from HHS emPOWER).
- **I – Independence:** people with a disability, people with cognitive difficulty (memory, decisions) and people age 65 and older.
- **S – Safety and support:** people age 65 and older living alone, children and people living in group quarters (in the needs table).
- **T – Transportation:** people in households without a car.

### County total vs. one community

The county total counts the people living inside the tsunami evacuation zone as hit at the same time, so it is an upper bound. Use the "Largest single community" table to plan for an event that hits one community.

### Map

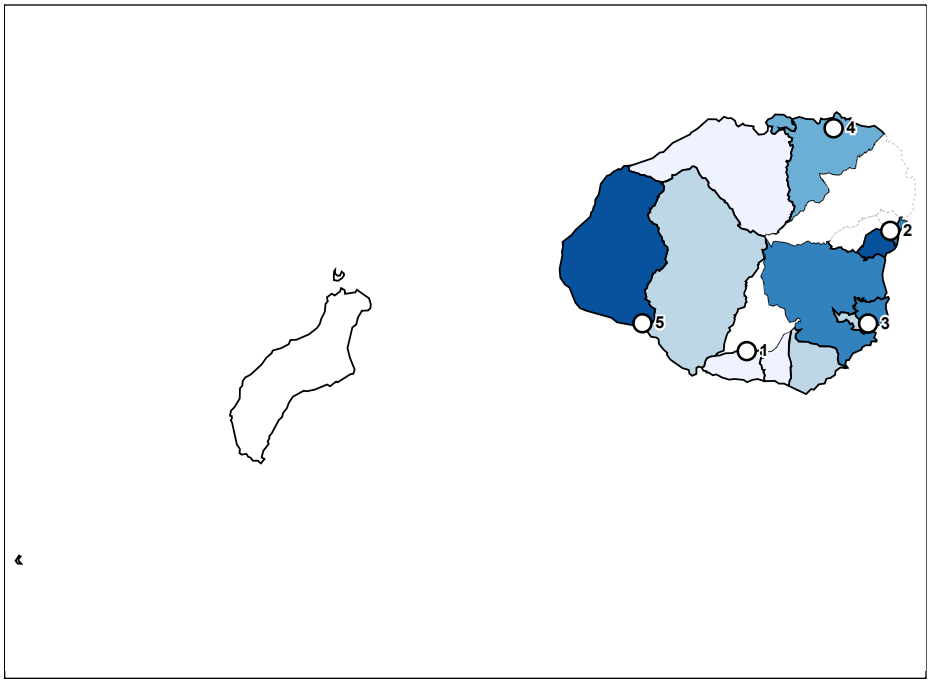
Darker blue means more people with needs, from "Lowest need" to "Highest need". A solid outline marks a Census area in the risk area; a dashed outline is outside it. Symbols: ○ usable site, △ check flood zone, ◇ flood risk not mapped, ⊠ not for this hazard. Numbers on the map match the site key.

### Why population numbers can differ

People in the risk area come from ACS 2018–2022 estimates (through the CDC Social Vulnerability Index 2022). Community names use 2020 Census counts. County totals in the plan use Census estimates for 2025. Each table names its source underneath.

**Every number is a planning estimate. In an event, follow county announcements.**

## 3. Map: people with needs and known sites




### Site key

**1** = Kalāheo Neighborhood Center (usable)
**2** = Kapa‘a Middle School (usable)
**3** = Kaua‘i High School Gym (usable)
**4** = Kīlauea Neighborhood Center (usable)
**5** = Waimea Canyon Middle School (usable)

- Lowest need (0–31 people)
- Lower (31–47 people)
- Middle (47–79 people)
- Higher (79–117 people)
- Highest need (117–211 people)
- Solid outline: Census area in the risk area
- Dashed outline: outside the risk area
- Usable site
- Check flood zone
- Flood risk not mapped (FEMA zone D)
- Not for this hazard

Color = people in the risk area who speak English less than well, use power-dependent medical equipment, have a disability, or live without a car (a person can be counted more than once). No street map: Census area outlines only.

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

**No 2026 activated site within a 15-minute drive (normal traffic)**, for sites usable for tsunami: 6 of 17 populated Census areas. With DOE candidate schools usable for this hazard: 5 of 17. This is not a statement that the area has no shelter: counties designate and open shelters for each event, and many public buildings are not on the 2026 list.

Drive-time matrix, per hazard: shortest drive from the area's land point to a 2026 activated site usable for that hazard on the same island; scenario 2 adds the 5 nearest DOE candidate schools usable for it. Site locations: U.S. Census Geocoder, checked against Esri ArcGIS World Geocoder. Travel times: Esri ArcGIS routing, normal traffic. Powered by Esri. Geocoding and routing: Esri ArcGIS Location Services.

## 4. Known sites (opened in 2026 events)

**Compiled from official county releases · verified by team**

| # | Status for this hazard | Site | Address | Why | Times opened in 2026 |
|---|---|---|---|---|---|
| 1 | ✓ **Usable** | Kalāheo Neighborhood Center | 4480 Papalina Road, Kalāheo | Outside the tsunami evacuation zone | 1 |
| 2 | ✓ **Usable** | Kapa‘a Middle School | 4867 Olohena Road, Kapa‘a | Outside the tsunami evacuation zone | 1 |
| 3 | ✓ **Usable** | Kaua‘i High School Gym | 3577 Lala Road, Līhu‘e | Outside the tsunami evacuation zone | 1 |
| 4 | ✓ **Usable** | Kīlauea Neighborhood Center | 2460 Keneke Street, Kīlauea | Outside the tsunami evacuation zone | 1 |
| 5 | ✓ **Usable** | Waimea Canyon Middle School | 9555 Huakai Road, Waimea | Outside the tsunami evacuation zone | 1 |

Source: county announcements of sites opened in 2026; FEMA flood zones (Hawai‘i Statewide GIS, State DFIRM); tsunami evacuation zones (Hawai‘i Statewide GIS).

Site locations: U.S. Census Geocoder, checked against Esri ArcGIS World Geocoder. Travel times: Esri ArcGIS routing, normal traffic. Powered by Esri. Geocoding and routing: Esri ArcGIS Location Services.

Candidate public schools in this county usable for this hazard: **15**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

### Care facilities in hazard zones

**Care facilities in hazard zones – Kaua‘i, Tsunami (8 of 36), all in: Tsunami evacuation zone**

- All Saints Preschool (Childcare centre)
- Aloha School Early Learning Center (Childcare centre)
- Kahili Adventist Preschool (Childcare centre)
- Kauai Independent Daycare Services, Inc.(K.I.D.S.) (Childcare centre)
- Ke Kula Ni‘ihau O Kekaha PCS (Childcare centre)
- Kekaha Elementary School (Childcare centre)
- Kekaha Head Start (Childcare centre)
- St Theresa Early Learning Academy (Childcare centre)

Source: CMS Provider Data Catalog (dialysis facilities, nursing homes, hospitals; facility addresses only), located with Esri ArcGIS World Geocoder; zones: State tsunami evacuation zone and FEMA flood zones (Hawai‘i Statewide GIS, State DFIRM).

Source: Ready Keiki Locator, all pre-K sites (Office of the Lieutenant Governor, published through Hawai‘i Statewide GIS): licensed group child care, public pre-K and charter pre-K sites only – no family child care homes; zones: State tsunami evacuation zone and FEMA flood zones (Hawai‘i Statewide GIS, State DFIRM).

## 5. Needs and proposed partners

Likely estimate. Roles are **proposed – to be confirmed with partner**. The pledged column stays blank until a partner confirms it.

| Need | How many | Unit | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Cots / sleeping space | 309 | cots | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Floor space (≤72 h) | 6,161 | sq ft | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Floor space (≤2 weeks) | 12,322 | sq ft | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Water | 309 | gal/day | American Red Cross | Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community | — | confirm with partner |
| Meals | 617 | meals/day | American Red Cross | Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community | — | confirm with partner |
| Toilets (≤72 h) | 8 | toilets | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Toilets (≤2 weeks) | 16 | toilets | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Shelter staff | 25 | staff per shift | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Office of Hawaiian Affairs (OHA), The Salvation Army, Philanthropy (foundations and donors), Business community | — | confirm with partner |
| Power outlets for power-dependent residents | 2 | outlets | Hawai‘i Department of Health (DOH) | County Emergency Management, Health care organizations (incl. managed health care), Utilities | — | confirm with partner |
| Accessible transport | 204 | people in no-vehicle households at risk | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Transportation (DOT), Business community, Other state and county agencies | — | confirm with partner |
| Pet space – dog households | 45 | households with a dog | County Emergency Management | Hawai‘i VOAD (Voluntary Organizations Active in Disaster) | — | confirm with partner |
| Pet space – cat households | 27 | households with a cat | County Emergency Management | Hawai‘i VOAD (Voluntary Organizations Active in Disaster) | — | confirm with partner |
| Children's area | 65 | children in shelter | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i Department of Health (DOH), Health care organizations (incl. managed health care), Philanthropy (foundations and donors) | — | confirm with partner |
| Interpreters – Ilocano | 3 | people with limited English in shelter | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Interpreters – Tagalog/Filipino | 2 | people with limited English in shelter | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Spanish | 1 | people with limited English in shelter | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Japanese | 0 | people with limited English in shelter | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Marshallese | 0 | people with limited English in shelter | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Hawaiian | 0 | people with limited English in shelter | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |

Source: likely estimate × planning factors (section 8; Florida Division of Emergency Management 2010; AVMA (2020 data) for pets).

**Service animals stay with their handler (ADA); not counted as pets.**

### Also plan for (county counts, not included in the totals above)

| Group | People | Detail | Date |
|---|---|---|---|
| People experiencing homelessness (Point-in-Time count) | 516 | sheltered 76, unsheltered 440; one-night count | 2026-01-25 |
| Visitors on island (average per day) | 28,577 | Kaua‘i | 2025 (annual average) |
| People in dam evacuation zones | 10,819 | State dam evacuation zones | zones fetched 2026-10-28 |

Source: Point-in-Time counts – Bridging the Gap 2026 report (Hawai‘i, Maui, Kaua‘i; night of Jan 25, 2026) and Partners In Care 2026 count for O‘ahu as released by the City and County of Honolulu; visitors – DBEDT State of Hawai‘i Data Book 2025, Table 7.06. Details: section 8.

Source: State of Hawai‘i dam evacuation zones (published through Hawai‘i Statewide GIS). People = share of each Census area inside a zone × its population, the same method as for the tsunami zone.

**Who else to plan for**

- **Domestic-violence survivors:** Served through confidential providers; no counts by design.
- **Off-grid households:** No public Census-area data; county and health partners to supply.
- **People on dialysis or receiving home care:** No public Census-area data; county and health partners to supply.
- **Earthquake:** Not modeled; listed as a limitation in the plan.
- **Caregivers and personal care attendants:** Stay with the person they support; count them for cots, meals and space.
- **Behavioral health:** Mental-health staff at each shelter (ratio: section 8).
- **Ni‘ihau:** Privately owned island, no public shelter; plan with the landowner and Kaua‘i County.
- **Families afraid to seek shelter (including immigrant families):** The county publishes who may use shelters; trusted community messengers share it.

## 6. Decision points

Proposed – supports, does not replace, county plans.

| When | Trigger | Action | Owner (proposed) | Support |
|---|---|---|---|---|
| **BEFORE** | Annual pre-positioning review (no official product)<br>County (sets the date) | PRE-POSITION: confirm shelters/refuge outside the tsunami evacuation zone (and extreme zone), transport plan for no-vehicle residents inside the zone, pre-approved multilingual evacuation messages. | County Emergency Management (each county) | Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Human Services (DHS), Hawai‘i Department of Transportation (DOT) |
| **BEFORE** | Before each season: who may use shelters (no official product)<br>County (sets the date) | County confirms and publishes who may use shelters and what ID is needed; trusted community messengers share it. | County Emergency Management (each county) | Hawai‘i Department of Human Services (DHS), Faith-based organizations, Office of Hawaiian Affairs (OHA), Hawai‘i VOAD (Voluntary Organizations Active in Disaster) |
| **DURING** | Tsunami Warning<br>NOAA Pacific Tsunami Warning Center (PTWC) | Evacuate the zone; open refuge outside the zone; prioritize residents who cannot evacuate on their own. | County Emergency Management (each county) | Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Transportation (DOT), American Red Cross |
| **AFTER** | Warning cancelled or all clear<br>County / NWS / PTWC | Shelter displaced households, feeding, power for medical devices, donations and volunteers. | American Red Cross | Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Hawai‘i Department of Health (DOH), Utilities, The Salvation Army |

Source: National Weather Service, Central Pacific Hurricane Center and Pacific Tsunami Warning Center product definitions (links: section 8).

## 7. Priority outreach and languages

### Top 10 Census areas by number of people with needs

| # | Community (Census area) | People in the risk area | Use power-dependent medical equipment (estimate) | People with a disability | Speak English less than well | People in households without a car | Needs total (a person can be in more than one column) |
|---|---|---|---|---|---|---|---|
| 1 | Kekaha / Waimea<br>Census area 409 | 1,159 | 6 | 92 | 31 | 82 | 211 |
| 2 | Wailua / Wailua Homesteads<br>Census area 402.05 | 964 | 6 | 103 | 8 | 14 | 131 |
| 3 | Līhuʻe<br>Census area 405 | 664 | 3 | 78 | 25 | 11 | 117 |
| 4 | Kapaʻa<br>Census area 403.02 | 1,369 | 7 | 97 | 12 | 0 | 116 |
| 5 | Hanamāʻulu / Puhi<br>Census area 404.01 | 360 | 1 | 35 | 24 | 23 | 82 |
| 6 | Princeville<br>Census area 401.06 | 401 | 3 | 63 | 3 | 10 | 79 |
| 7 | Kīlauea<br>Census area 401.05 | 338 | 2 | 39 | 7 | 6 | 54 |
| 8 | Līhuʻe / Puhi<br>Census area 404.02 | 173 | 1 | 26 | 4 | 15 | 47 |
| 9 | Hanapēpē<br>Census area 408 | 135 | 1 | 18 | 15 | 5 | 39 |
| 10 | Kōloa / Poʻipū<br>Census area 406.03 | 166 | 1 | 16 | 3 | 14 | 34 |

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

### More needs in the largest communities

| Community (Census area) | Hearing difficulty | Vision difficulty | Cognitive difficulty (memory, decisions) | Age 65 and older, living alone |
|---|---|---|---|---|
| Kapaʻa<br>Census area 403.02 | 33 | 17 | 96 | 63 |
| Kekaha / Waimea<br>Census area 409 | 18 | 15 | 30 | 55 |
| Wailua / Wailua Homesteads<br>Census area 402.05 | 46 | 26 | 71 | 47 |

Source: ACS 2020–2024 5-year Census-area tables B18102 (hearing), B18103 (vision), B18104 (cognitive) and B09020 (age 65 and older living alone), times the share of each area in the risk area; homes in 10+ unit buildings: CDC/ATSDR SVI 2022 (E_MUNIT, housing units).

### How to reach residents

| Channel | Owner (proposed) | Support (proposed) |
|---|---|---|
| Official alerts: Wireless Emergency Alerts / Emergency Alert System and county alert sign-ups | County Emergency Management (each county) | Hawaiʻi Emergency Management Agency (HI-EMA) |
| Broadcast radio, including in-language programs where available | County Emergency Management (each county) | Hawaiʻi Emergency Management Agency (HI-EMA) |
| Outdoor warning sirens: a signal to turn on radio, TV or phone alerts for instructions | Hawaiʻi Emergency Management Agency (HI-EMA) | County Emergency Management |
| Trusted community messengers: faith-based groups, Native Hawaiian organizations, health care providers, community nonprofits | Hawaiʻi Department of Human Services (DHS) | Faith-based organizations, Office of Hawaiian Affairs (OHA), Health care organizations (incl. managed health care), Hawaiʻi VOAD (Voluntary Organizations Active in Disaster) |
| Printed notices at shelters, libraries and schools, in the county's top languages | County Emergency Management (each county) | Hawaiʻi State Public Library System, Hawaiʻi Department of Education (DOE) – schools |

Source: proposed channels and formats in the plan's communication pathways – supports, does not replace, county and State public-information plans.

**Populated Census areas with 0 outdoor sirens (2): prioritise WEA, radio and trusted messengers.** Kapaʻa (403.01); Princeville (401.06)

Source: Hawaiʻi Emergency Management Agency public siren map (locations of permanent outdoor warning sirens; no live status). Audible radius: 3,400 feet, the manufacturer rating published by the Hawaiʻi Emergency Management Agency; the real range varies with terrain, buildings and weather.

**Accessible formats for every message:**

- **ASL video:** short American Sign Language video of each message
- **Captions:** captions on every video and broadcast
- **Large print:** printed notices in large print
- **Plain language:** short sentences, one action at a time
- **Screen-reader-friendly text:** plain text or tagged web pages, not images of text

### Languages: people age 5 and older who speak English less than "very well" (Kauaʻi + Maui)

Kauaʻi and Maui share one language list: the Census survey reports both counties (and Kalawao) as one area, so they cannot be separated.

| Language | People | Margin of error (90%) |
|---|---|---|
| Ilocano | 9,131 | ±1,439 |
| Tagalog/Filipino | 5,074 | ±926 |
| Spanish | 1,711 | ±554 |
| Japanese | 1,339 | ±417 |
| Marshallese | 633 | ±488 |
| Hawaiian | 515 | ±289 |

Source: ACS PUMS 2019–2023 5-year (validated against the DBEDT State of Hawaiʻi Data Book 2024, Table 1.44).

Pre-approved message templates: none yet; no machine translation without human review.

## 8. Sources and assumptions

### Planning factors

| Factor | Value | Source or team assumption |
|---|---|---|
| Shelter-seeking rate – Hurricane | 2% / 3% / 5% | South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%) |
| Shelter-seeking rate – Wildfire | 2% / 5% / 11.2% | high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are Team assumption. |
| Shelter-seeking rate – Tsunami | 2% / 5% / 11% | Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Shelter-seeking rate – Inland flooding | 2% / 5% / 11% | Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Shelter-seeking rate – Coastal flooding | 2% / 5% / 11% | Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Footprint – Hurricane | Every Census area | Hurricanes affect whole islands (Lala, Lowell 2026). |
| Footprint – Tsunami | Share living inside the tsunami evacuation zone | Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately. |
| Footprint – Wildfire | Top 25% of Census areas by FEMA risk score | Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption). |
| Footprint – Inland flooding | Top 25% of Census areas by FEMA risk score | Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption). |
| Footprint – Coastal flooding | Top 25% of Census areas by FEMA risk score | Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption). |
| Floor sqft 72h | 20 | Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h |
| Floor sqft 2wk | 40 | Florida DEM (2010): short-term shelter, <=2 weeks |
| Water gal per day | 1 | Florida DEM (2010) |
| Meals per day | 2 | Florida DEM (2010): two hot meals a day plus snacks |
| Persons per toilet short term | 20 | Florida DEM (2010): short-term shelter, <=2 weeks |
| Power outlets per power dependent person | 1 | Team assumption |
| Persons per toilet 72h | 40 | Florida DEM (2010): emergency evacuation shelter, <=72 h |
| Staffing ratios (one staff member per N residents) | client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12 | One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations. |
| Dog owning household share | 0.446 | AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27. |
| Cat owning household share | 0.26 | AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'. |

### Data sources

| Source | Fetched | URL / file |
|---|---|---|
| CDC/ATSDR Social Vulnerability Index 2022, census tracts | 2026-10-27 | https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2 |
| FEMA National Risk Index v1.20, census tracts – Hurricane | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Ind |

| Source | Fetched | URL / file |
|---|---|---|
| FEMA National Risk Index v1.20, census tracts – Wildfire | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index |
| FEMA National Risk Index v1.20, census tracts – Tsunami | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Inde |
| FEMA National Risk Index v1.20, census tracts – Inland Flooding | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Ri |
| FEMA National Risk Index v1.20, census tracts – Coastal Flooding | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_F |
| Tsunami Evacuation Zones (Statewide GIS) | 2026-10-27 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2 |
| Extreme Tsunami Evacuation Zones (Statewide GIS) | 2026-10-27 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12 |
| HHS emPOWER – electricity-dependent DME, county level | 2026-10-27 | https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2 |
| ACS 2020-2024 5-year, tract tables B18102 (hearing), B18103 (vision), B18104 (cognitive), B09020 (age 65+ living alone) | 2026-10-26 | https://data.census.gov/api/access/data/table?g=040XX00US15%241400000&id=ACSDT5Y2024.B18102 |
| Hawai‘i Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26) | 2026-10-28 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6 |
| County shelter activations 2026 (official county releases) | 2026-10-27 | County shelter activation table: one official county source per row |
| 2020 Census Block Assignment File, place/CDP (tract community labels) | 2026-10-27 | https://www2.census.gov/geo/docs/maps-data/data/baf2020/BlockAssign_ST15_HI.zip |
| 2020 Census PL 94-171 block population P1_001N (tract community labels) | 2026-10-27 | https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/Hawaii/hi2020.pl.zip |
| ACS PUMS 2019-2023 5-year (detailed languages) | 2026-10-27 | https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip |

| Source | Fetched | URL / file |
|---|---|---|
| DBEDT State of Hawai‘i Data Book 2024, Table 1.44 (validation) | 2026-10-27 | https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf |
| Partner list: Hawai‘i DHS – ESF-6 challenge statement (2026) | 2026-10-15 | Hawai‘i DHS – ESF-6 challenge statement (2026) |
| pre season hurricane | | https://www.nhc.noaa.gov/cphcmission.php |
| hurricane watch | | https://www.nhc.noaa.gov/aboutgloss.shtml |
| hurricane warning | | https://www.nhc.noaa.gov/aboutgloss.shtml |
| tsunami warning | | https://www.tsunami.gov/?page=message_definitions |
| flash flood warning | | https://www.weather.gov/safety/flood-watch-warning |
| red flag warning | | https://www.weather.gov/safety/wildfire-ww |
| coastal flood warning | | https://www.weather.gov/okx/marinedef |
| coastal flood warning in effect | | https://www.weather.gov/okx/marinedef |
| Ready Keiki Locator – All Pre-K Sites (Office of the Lieutenant Governor, hosted by Hawai‘i Statewide GIS) | 2026-10-28 | https://services.arcgis.com/HQ0xoN0EzDPBOEci/arcgis/rest/services/Ready_Keiki_GIS/FeatureServer/0 |
| Hawai‘i Emergency Management Agency – all-hazard outdoor warning sirens (public siren layer; locations only) | 2026-10-28 | https://services9.arcgis.com/aKxrz4vDVjfUwBWJ/arcgis/rest/services/public_siren_status_layer_view/FeatureServer/0 |
| State of Hawai‘i dam evacuation zones (hosted by Hawai‘i Statewide GIS) | 2026-10-28 | https://services.arcgis.com/HQ0xoN0EzDPBOEci/arcgis/rest/services/DamEvacZones_joined/FeatureServer/0 |
| Hawai‘i Emergency Management Agency – siren audible radius (3,400 feet, manufacturer rating) | 2026-10-28 | https://dod.hawaii.gov/hiema/all-hazard-statewide-outdoor-warning-siren-system/ |
| CMS Provider Data Catalog – Dialysis Facility – Listing by Facility | | https://data.cms.gov/provider-data/dataset/23ew-n7w9 |
| CMS Provider Data Catalog – Nursing Homes – Provider Information | | https://data.cms.gov/provider-data/dataset/4pq5-n9py |
| CMS Provider Data Catalog – Hospital General Information | | https://data.cms.gov/provider-data/dataset/xubh-q36u |

| Source | Fetched | URL / file |
|---|---|---|
| Esri ArcGIS World Geocoder (site check, care facilities) and ArcGIS routing (drive and walk times, normal traffic). Powered by Esri. | | https://developers.arcgis.com/rest/ |

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.

## 9. Results

- **Coverage:** 436 of 436 Census areas have needs data (FEMA risk data joined for 435 of 436). Source: coverage check of the needs layer (CDC SVI 2022, FEMA NRI v1.20).
- **Backtest – Lahaina wildfire (2023-08-11):** observed 1,418 people in shelters; model 254 / 635 / 1,423 (low / likely / high). This is a calibration event, not validation: the high rate was set from it. Source: Lahaina 2023 backtest (Maui Now count of people in shelters).
- **Time:** a full rebuild from local data (14 steps, 20 scenarios, 20 playbooks) took 71.7 seconds on Apple M4 (2026-10-28). Team-assumed baseline without this tool: 2 working days per county per event. Sources: timed full rebuild from local data; Team assumption, to be confirmed with DHS in a timed tabletop exercise.